

Varsha Hannah George



Genetics & Molecular Biology | Research & Bioinformatics

Profile

Varsha Hannah George is a research-oriented **Genetics and Molecular Biology professional** with strong experience in molecular biology, microbiome research, next-generation sequencing, bioinformatics, vaccine research, and mitochondrial biology.

A graduate of the **School of Biomedical Sciences, The Apollo University**, Varsha has developed a strong interdisciplinary research profile by combining experimental laboratory techniques with computational and data-driven approaches. Her work focuses on understanding molecular mechanisms underlying health and disease and applying modern biological tools to address complex research questions.

She completed her **B.Sc. (Hons) in Genetics and Molecular Biology with an outstanding CGPA of 9.64**, reflecting her strong academic performance and dedication to biomedical sciences.

Academic Journey

B.Sc. (Hons) in Genetics and Molecular Biology

School of Biomedical Sciences, The Apollo University, Chittoor
2022–2026 | **CGPA: 9.64/10**

Senior Secondary Education – Biology, Physics & Chemistry

Board of Intermediate Education, Andhra Pradesh
2019–2021 | **92.5%**

Research Experience

Undergraduate Honours Research Thesis – The Apollo University

Varsha's undergraduate research focused on **gut microbial metabolites and their functional roles in disease**. Her work involved microbiome analysis, pathway inference, statistical evaluation, and computational approaches to investigate **gut microbe–host interactions** and understand microbiome-associated biological patterns.

Research Internship – IIT Tirupati

During her summer research internship at **IIT Tirupati**, she contributed to **carbohydrate-based vaccine research**, with a focus on antigen design and conjugation strategies. She also gained practical experience in column chromatography and thin-layer chromatography for the analysis of carbohydrate derivatives, strengthening her understanding of glycol-immunology.

Research Training – Sri Venkateswara University

Her research training included **recombinant DNA techniques**, bacterial transformation, gene-expression workflows, plant tissue culture, regeneration, and tissue differentiation. This provided her with valuable exposure to both molecular biology and plant biotechnology.

Research Internship – Dr. Reddy's Institute of Life Sciences

At Dr. Reddy's Institute of Life Sciences, Varsha worked on **mitochondrial research using HEK cells**, including mitochondrial isolation and assessment of cytochrome c oxidase (Complex IV) activity. She also developed hands-on proficiency in **qPCR, Western blotting, protein quantification, and molecular and biochemical laboratory techniques**.

Publications & Scholarly Contributions

Varsha has already established an impressive publication record during her academic journey, contributing to research in **cancer biology, molecular modelling, phytochemical research, and metabolic engineering**.

Selected Publications

- **Modelling the Effect of Docosa Hexaenoic Acid (DHA) in Cancer Therapy** – published in *Onkologia i Radioterapia* (2024), a Scopus Q1 journal.
- **Heptadecanol: A phytochemical multi-target inhibitor of SMYD3 & GFPT2 proteins in non-small cell lung cancer** – published in *Journal of Computer-Aided Molecular Design* (2025), a Scopus Q2 journal.
- **Future Perspectives and Emerging Trends** – contributed a book chapter to *Novel Bioproducts through Metabolic Engineering of Secondary Metabolites*, published by Academic Press, Elsevier (2026).

She has also contributed to a manuscript under review on the **in-silico and in-vitro evaluation of rose oxide as a multi-target inhibitor against *Acinetobacter baumannii***.

Research & Technical Expertise

Her laboratory and computational expertise includes:

- Molecular biology and cellular techniques
- PCR and qPCR
- DNA/RNA isolation
- Gel electrophoresis
- ELISA
- Western blotting and SDS-PAGE
- Mitochondrial isolation and biochemical assays
- Next-generation sequencing workflows
- DNA library preparation and sequencing analysis
- Microbiome analysis
- Bioinformatics and pathway prediction
- Statistical analysis and scientific visualization
- QIIME2, PICRUSt2 and R-based analysis
- Chromatography and laboratory instrumentation

Awards & Recognition

Varsha's academic excellence has been recognized through several notable achievements:

- **Gold Medalist – NPTEL Bioreactor Certification (2024)**
- **Silver Medalist – NPTEL Classics in Neuroscience (2025)**
- **Silver Medalist – NPTEL Comprehensive Molecular Diagnostics and Advanced Gene Expression Analysis (2025)**
- **Bronze Medal – National Science Olympiad, Science Olympiad Foundation**
- **Participant, 11th International Conference on Transforming Healthcare with AI**

Research Interests

Her interdisciplinary research interests include:

- Microbiome and gut–host interactions
- Molecular mechanisms of disease
- Cancer biology and therapeutics
- Molecular diagnostics
- Bioinformatics and computational biology
- Next-generation sequencing
- Vaccine research
- Mitochondrial biology

- Molecular and biochemical research

An Inspiring Journey

Varsha Hannah George represents the spirit of **curiosity, scientific excellence, and interdisciplinary research**. Her journey from an undergraduate student in Genetics and Molecular Biology to an active contributor to scientific publications and advanced research projects demonstrates the opportunities that emerge when strong academic foundations are combined with hands-on research experience.

Her achievements across **microbiology, molecular biology, bioinformatics, cancer research, vaccine research, and computational biology** reflect a versatile and research-driven approach to biomedical science.

Her journey is an inspiring example for students of the **Department of Biomedical Sciences, The Apollo University**, showing how dedication, curiosity, and early exposure to research can open pathways toward meaningful contributions to modern life sciences.